Lesson 4: Input — Talking to the Computer ⌨️

> **Goal:** Use `input()` to read data from the user, convert types, and build a simple calculator.

---

1. What is Input?

So far, all your programs had fixed values — you typed the numbers directly in the code. But real programs ask the **user** for information!

In Scratch, you used the **"ask and wait"** block to get input from the user. In Python, we use the `input()` function.

Scratch	Python
"ask and wait" block	`input()`
"answer" variable	the value returned by `input()`

---

2. Basic `input()`

```
name = input("What is your name? ")
print("Hello,", name)
```

How it works:

1. Python shows the message `What is your name?` on the screen.
2. The program **pauses** and waits for you to type something.
3. When you press **Enter**, Python stores what you typed in the variable `name`.
4. `print()` shows the greeting.

Example run:

```
What is your name? An
Hello, An
```

The text inside `input("...")` is called a **prompt** — it tells the user what to type.

---

3. `input()` Always Returns a String

This is very important! No matter what the user types, `input()` always gives you a **string** (text).

```
x = input("Enter a number: ")
print(x)
print(type(x))
```

Example run:

```
Enter a number: 42
42
<class 'str'>
```

Even though the user typed `42`, Python stores it as the string `"42"`, not the number `42`. If you try to do math with it, you get an error!

```
x = input("Enter a number: ")
print(x + 10)    # ❌ Error! Can't add string and number
```

---

4. Type Conversion — Changing the Type

To use the input as a number, you must **convert** it. This is called **type conversion** (or **casting**).

Function	Converts to	Example
`int()`	Integer (whole number)	`int("42")` → `42`
`float()`	Float (decimal number)	`float("3.14")` → `3.14`
`str()`	String (text)	`str(42)` → `"42"`

Reading an integer:

```
age = int(input("Enter your age: "))
print("Next year you will be", age + 1)
```

Example run:

```
Enter your age: 10
Next year you will be 11
```

Reading a float:

```
height = float(input("Enter your height in meters: "))
print("Your height:", height)
```

Example run:

```
Enter your height in meters: 1.35
Your height: 1.35
```

---

5. How Type Conversion Works

Think of it like a machine that transforms one thing into another:

```
"42"  →  int()  →  42
"3.14"  →  float()  →  3.14
42  →  str()  →  "42"
```

You can also convert between numbers:

```
x = int(3.9)     # x = 3  (cuts off the decimal, does NOT round)
y = float(5)     # y = 5.0
```

> ⚠️ `int()` does NOT round — it just removes the decimal part. `int(3.9)` gives `3`, not `4`.

---

6. Reading Multiple Inputs

You can call `input()` multiple times to read several values:

```
first_name = input("First name: ")
last_name = input("Last name: ")
print("Full name:", first_name, last_name)
```

Example run:

```
First name: Nguyen
Last name: An
Full name: Nguyen An
```

---

7. Build a Calculator

Now let's put it all together and build a simple calculator!

```
# Simple calculator
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Sum:", a + b)
print("Difference:", a - b)
print("Product:", a * b)
print("Quotient:", a / b)
```

Example run:

```
Enter first number: 10
Enter second number: 4
Sum: 14.0
Difference: 6.0
Product: 40.0
Quotient: 2.5
```

---

8. Common Mistakes

```
# Mistake 1: Forgetting to convert
age = input("Age: ")
print(age + 1)          # ❌ TypeError: can only concatenate str to str

# Fix:
age = int(input("Age: "))
print(age + 1)          # ✅

# Mistake 2: Using int() for a decimal
height = int(input("Height: "))   # ❌ If user types 1.75, this crashes!

# Fix:
height = float(input("Height: "))  # ✅

# Mistake 3: Forgetting the prompt message
x = input()             # ✅ Works, but user doesn't know what to type
x = input("Enter x: ")  # ✅ Better — tells the user what to do
```

---

9. Vocabulary

English	Vietnamese	Meaning
input	đầu vào	Data given to the program by the user
output	đầu ra	Data shown by the program
prompt	lời nhắc	The message shown before waiting for input
type conversion	chuyển đổi kiểu	Changing data from one type to another
integer	số nguyên	A whole number (no decimal point)
float	số thực	A number with a decimal point
string	chuỗi	Text data, written in quotes
cast	ép kiểu	Another word for type conversion
user	người dùng	The person running the program
standard input	đầu vào chuẩn	The keyboard input stream
return	trả về	What a function gives back
convert	chuyển đổi	Change from one form to another

---

10. Summary

✅ `input()` pauses the program and waits for the user to type something. ✅ `input()` **always returns a string** — even if the user types a number. ✅ Use `int(input())` to read a whole number. ✅ Use `float(input())` to read a decimal number. ✅ The text inside `input("...")` is the prompt — it tells the user what to type. ✅ `int()`, `float()`, `str()` convert between types.

---

11. Homework

🟢 Easy

Exercise E1. Say My Name

Ask the user for their name and print a greeting.

- **Input:** One line — the user's name.
- **Output:** One line — a greeting with the name.

Example:

Input	Output
An	Hello, An!
Minh	Hello, Minh!

Exercise E2. Double a Number

Ask the user for a whole number. Print the number doubled.

- **Input:** One integer.
- **Output:** The number multiplied by 2.

Example:

Input	Output
5	10
13	26

Exercise E3. Sum of Two Numbers

Ask the user for two whole numbers (one per line). Print their sum.

- **Input:** Two integers, one per line.
- **Output:** Their sum.

Example:

Input	Output
3 / 7	10
100 / 200	300

Exercise E4. Name and Age

Ask the user for their name and age. Print a sentence combining both.

- **Input:** Two lines — name, then age.
- **Output:** One sentence.

Example:

Input	Output
An / 10	An is 10 years old.
Minh / 8	Minh is 8 years old.

Exercise E5. Square Area

Ask the user for the side length of a square. Print the area.

- **Input:** One integer — the side length.
- **Output:** The area (side × side).

Example:

Input	Output
5	25
12	144

Exercise E6. Multiply Three Numbers

Ask the user for three whole numbers (one per line). Print their product.

- **Input:** Three integers, one per line.
- **Output:** Their product.

Example:

Input	Output
2 / 3 / 4	24
5 / 5 / 5	125

Exercise E7. Greeting with City

Ask the user for their name and their city. Print a sentence like: `Hi An, welcome from Ho Chi Minh City!`

- **Input:** Two lines — name, then city.
- **Output:** One greeting sentence.

Example:

Input	Output
An / Ho Chi Minh City	Hi An, welcome from Ho Chi Minh City!
Minh / Hanoi	Hi Minh, welcome from Hanoi!

🟡 Medium

Exercise M1. Average of Three

Ask the user for three decimal numbers. Print their average.

- **Input:** Three floats, one per line.
- **Output:** Their average.

Example:

Input	Output
7.0 / 8.0 / 9.0	8.0
5.5 / 6.5 / 7.0	6.333333333333333

Exercise M2. Rectangle

Ask the user for the width and height of a rectangle. Print the area and perimeter.

- **Input:** Two integers — width, then height.
- **Output:** Two lines: area, then perimeter.

Example:

Input	Output
6 / 4	24
	20
10 / 3	30
	26

Exercise M3. Celsius to Fahrenheit

Ask the user for a temperature in Celsius. Print the temperature in Fahrenheit. Formula: `F = C * 9 / 5 + 32`.

- **Input:** One float — temperature in Celsius.
- **Output:** Temperature in Fahrenheit.

Example:

Input	Output
0	32.0
100	212.0
37	98.6

Exercise M4. Seconds to Minutes

Ask the user for a number of seconds. Print how many full minutes and remaining seconds.

- **Input:** One integer — total seconds.
- **Output:** Two lines: full minutes, then remaining seconds.

Example:

Input	Output
90	1
	30
125	2
	5

Exercise M5. Circle

Ask the user for the radius of a circle. Print the area and circumference. Use `pi = 3.14159`.

- **Input:** One float — the radius.
- **Output:** Two lines: area, then circumference.

Example:

Input	Output
5	78.53975
	31.4159

Exercise M6. Tip Calculator

Ask the user for a bill amount and a tip percentage. Print the tip amount and the total bill.

- **Input:** Two floats — bill amount, then tip percentage.
- **Output:** Two lines: tip amount, then total.

Example:

Input	Output
100000 / 10	10000.0
	110000.0
200000 / 15	30000.0
	230000.0

Exercise M7. Power

Ask the user for a base number and an exponent. Print the result of base raised to the exponent. Use the `**` operator.

- **Input:** Two integers — base, then exponent.
- **Output:** The result.

Example:

Input	Output
2 / 10	1024
3 / 4	81

🔴 Hard

Exercise H1. Age Calculator

Hoa is filling out a form and needs to know her age. She knows the year she was born, but not her current age. The current year is **2025**.

Write a program that reads the birth year and prints the person's age.

- **Input:** One integer — the birth year (1900 ≤ year ≤ 2025).
- **Output:** One integer — the age in 2025.

Example:

Input	Output
2015	10
2000	25
1990	35

Explanation: Age = 2025 – birth year. If someone was born in 2015, they are 2025 – 2015 = 10 years old.

Exercise H2. Rectangle Dimensions

A builder needs to quickly calculate the area and perimeter of a rectangular room. Help him by writing a program that reads the width and height, then prints both values.

- **Input:** Two integers on separate lines — width W, then height H (1 ≤ W, H ≤ 1000).
- **Output:** Two lines: area on line 1, perimeter on line 2.

Example:

Input	Output
8 / 5	40
	26
12 / 7	84
	38

Explanation: Area = W × H. Perimeter = 2 × (W + H).

Exercise H3. Seconds Converter

An astronaut on a space station needs to convert mission time from total seconds into hours, minutes, and seconds for the mission log.

Write a program that reads a total number of seconds and prints the equivalent hours, minutes, and seconds.

- **Input:** One integer — total seconds (0 ≤ seconds ≤ 86400).
- **Output:** Three lines: hours, then minutes, then seconds.

Example:

Input	Output
3661	1
	1
	1
7200	2
	0
	0
90	0
	1
	30

Explanation: 3661 seconds = 1 hour (3600s) + 1 minute (60s) + 1 second. Use `//` and `%`.

Exercise H4. Coin Change

A vending machine only has coins of 500 dong, 200 dong, and 100 dong. When a customer pays too much, the machine gives change using the **fewest coins possible** — starting with the largest coins first.

Write a program that reads an amount (a multiple of 100) and prints how many of each coin to give.

- **Input:** One integer — the amount in dong (100 ≤ amount ≤ 10000, always a multiple of 100).

- **Output:** Three lines: number of 500-dong coins, number of 200-dong coins, number of 100-dong coins.

Example:

Input	Output
1300	2
	1
	1
800	1
	1
	1

Explanation: For 1300 dong: 2 × 500 = 1000, remainder 300. 1 × 200 = 200, remainder 100. 1 × 100 = 100. Done.

Exercise H5. BMI Calculator

A school nurse wants a quick tool to calculate students' Body Mass Index (BMI). BMI = weight (kg) ÷ (height (m) × height (m)).

Write a program that reads weight and height, then prints the BMI rounded to 2 decimal places.

- **Input:** Two lines — weight in kg (float), then height in meters (float).
- **Output:** One line — the BMI rounded to 2 decimal places.

Example:

Input	Output
30 / 1.35	16.46
50 / 1.60	19.53
70 / 1.75	22.86

Explanation: BMI = weight / (height * height). Use `round(bmi, 2)` to round to 2 decimal places.

Exercise H6. Speed, Distance, Time

A race car driver wants to know how long it will take to finish a race. He knows the total distance of the track and his average speed.

Write a program that reads the distance (km) and speed (km/h), then prints the time in hours.

- **Input:** Two lines — distance in km (float), then speed in km/h (float).
- **Output:** One line — the time in hours.

Example:

Input	Output
300 / 150	2.0
100 / 80	1.25
500 / 200	2.5

Explanation: Time = Distance ÷ Speed.

Exercise H7. Temperature Converter

A scientist needs to convert temperatures between different scales. She wants to convert from Celsius to both Fahrenheit and Kelvin at the same time.

Write a program that reads a temperature in Celsius and prints the equivalent in Fahrenheit and Kelvin.

- **Input:** One float — temperature in Celsius.
- **Output:** Two lines: Fahrenheit on line 1, Kelvin on line 2.

Formulas:

- Fahrenheit = Celsius × 9 / 5 + 32
- Kelvin = Celsius + 273.15

Example:

Input	Output
0	32.0
	273.15
100	212.0

Input	Output
	373.15
-40	-40.0
	233.15